

Adroit Technologies

Protocol Driver

Name	Modbus RTU/ASCII Serial
DLL	MODBUSI



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Contents

1. Overview	4
1.1. Modbus Emulators.....	4
2. Wiring	5
2.1. Wiring	5
3. Device Configuration	6
3.1. Device Configuration	6
3.1.1. PLC Details.....	6
3.1.2. Local Details.....	6
3.1.3. Handshaking	7
3.1.4. Network Latency/Delays	7
3.2. Modem Support	8
3.2.1. Use Dialup Modem.....	9
3.2.2. POLAR 3G Modem Support	10
3.3. Advanced Settings.....	11
3.3.1. Backoff	11
3.3.2. Byte Ordering	11
3.3.3. General.....	12
3.3.4. PDU lengths (see section: <i>Protocol unit sizes</i>)	12
3.3.5. Protocol Unit Sizes	13
4. Supported Addresses	14
4.1. Using Normal Modbus Type Addressing.	14
4.2. Using IEC Addressing :	16
5. Protocol Definition and Considerations	17
5.1. Diagnostic Sub-Functions.....	17
5.1.1. Function 08 Supported Addresses.....	17
5.1.2. Startup Function	17
5.1.3. Comms Recovery	17
5.1.4. Tag Configuration	18
5.1.5. Function Reference	18
5.1.6. Function 08 Registry Entries	20
5.2. Broadcast Support.....	20
5.3. Telemechanique and other support	21
5.3.1. GE Multilin Features	21
5.3.2. SuperCon Power Meters	21
5.3.3. MOSCAD	21
5.3.4. ABB Controllers.....	21
5.4. Advanced Registry Key Settings	22
5.5. Message Protocol	23
5.5.1. Coils 000001- 065535 (%Q0-%Q65535)	23
5.5.2. Inputs 100001- 165535 (%I0-%I65535).....	23
5.5.3. Input Registers 300001- 365535 (%IW0-%IW65535)	23
5.5.4. Holding Registers 400001- 465535 (%MW0-%MW65535)	24
5.5.5. Diagnostic 0x08 functions.....	24
5.5.6. Extended Registers 600000- 698303 (NB 600000 is valid) (%ME0-%ME98303)	25
6. General Notes and Troubleshooting	26
6.1. Contact Details.....	26
6.2. Adroit Technologies Knowledge Base.....	26
6.3. PLC Exception Codes	26
6.4. Trouble Shooting.....	26
7. Revisions.....	27
7.1. Document Revision.....	27
7.2. DLL Revision	27

1. Overview

Adroit supports communication with the following PLCs: Modicon 84, 884, 184, 384, 484, 584 and 984 amongst other devices using Modbus RTU and ASCII protocol.

1.1. Modbus Emulators

A number of hardware manufacturers have adopted the Modbus protocol to communicate to their hardware. These emulations have different maximum number of points that can be retrieved in one scan, amongst other variations. Since Adroit creates scan tasks/jobs automatically for you, it is important to choose a PLC type that would not result in a scan task requesting more data in one scan than that which can be handled by the device. Consult the list below to choose the most appropriate PLC model or select the “Custom” PLC type and supply your own from vendor documentation. See section: *Protocol unit sizes*.

2. Wiring

2.1. Wiring

PLC 25 pin		PC 25 pin
1	— Shield	1
2	—	3
3	—	2
4	] [	4
5		5
6	] [	6
8		8
20		20
7	—	7

9pin to 9pin.

Also used for ABB AC-400 series DCS (diagram is same as above, only it uses the 9 pin plug)

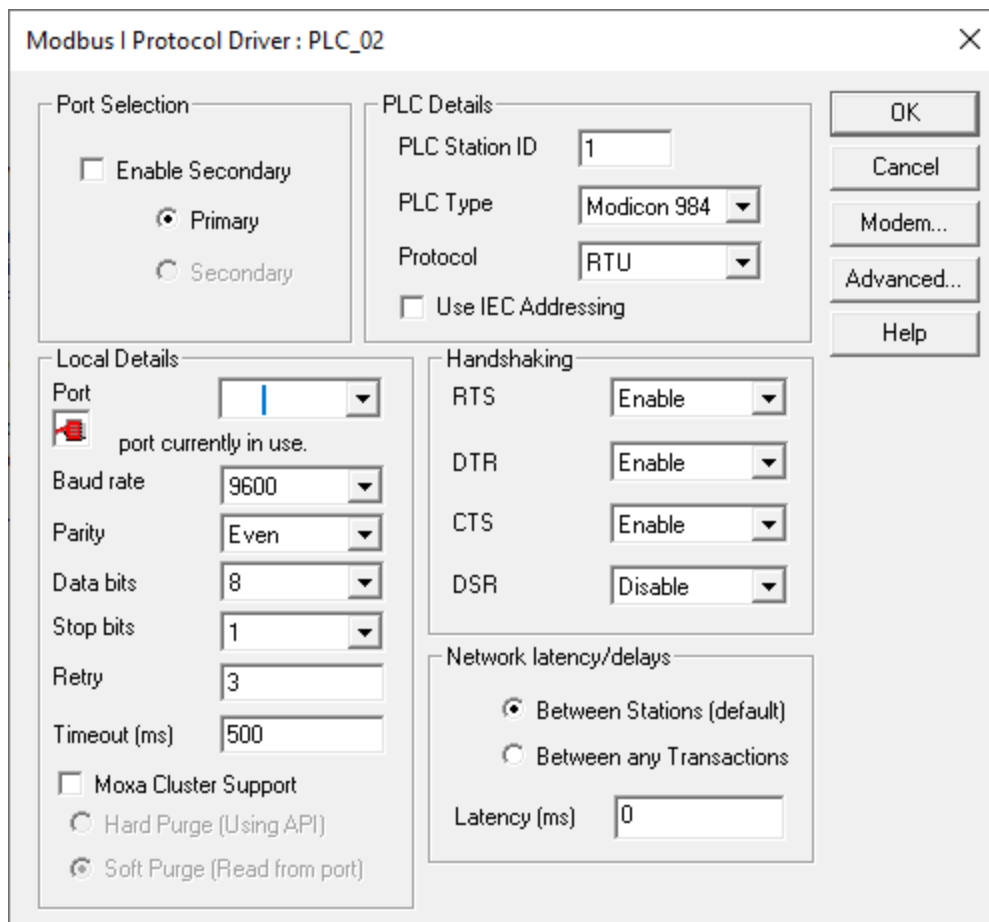
PLC 9 pin		PC 9 pin
2	—	3
3	—	2
5	—	5
7	] [	7
8		8
1	] [	1
4		4
6		6

Note: Connect shield to PLC side only

See section on Telemechanique and other for wiring diagrams to other devices.

3. Device Configuration

3.1. Device Configuration



The image shows a configuration window titled "Modbus I Protocol Driver : PLC_02". It contains several sections for configuring the PLC connection:

- Port Selection:** Includes checkboxes for "Enable Secondary", "Primary" (selected), and "Secondary".
- PLC Details:** Includes fields for "PLC Station ID" (1), "PLC Type" (Modicon 984), "Protocol" (RTU), and a checkbox for "Use IEC Addressing".
- Local Details:** Includes a "Port" dropdown (showing a red X icon and "port currently in use."), "Baud rate" (9600), "Parity" (Even), "Data bits" (8), "Stop bits" (1), "Retry" (3), "Timeout (ms)" (500), and checkboxes for "Moxa Cluster Support", "Hard Purge (Using API)", and "Soft Purge (Read from port)".
- Handshaking:** Includes dropdowns for "RTS" (Enable), "DTR" (Enable), "CTS" (Enable), and "DSR" (Disable).
- Network latency/delays:** Includes radio buttons for "Between Stations (default)" (selected) and "Between any Transactions", and a "Latency (ms)" field (0).

Buttons on the right include "OK", "Cancel", "Modem...", "Advanced...", and "Help".

3.1.1. PLC Details

PLC Station ID - Valid data highway station number.

PLC Type - Applicable Modicon PLC types (see: *Protocol Unit Sizes*)

Protocol - RTU, ASCII.

Use IEC Addressing – Instead of the Standard Modbus Addressing like 400001 it will display %MW0. (So %MW0 = 400001 = INDEX0 in the protocol).

3.1.2. Local Details

Port - The available serial ports on your machine. (This list is now refreshed in the background to cater for Ethernet extender ports, which are tested online. During this time, the port selection and OK is disabled.)

Baud Rate – The speed of data transmission used. Any of the following Baud Rates may be specified: 110, 300, 600, 1200, 2400, 4800, 9600, 19200, 64800, (115200, 128000 or 256000 if your UART permits).

Parity – The parity bit setting used in the data transmission. This can be None, Odd or Even.

Data bits – (Default = 8) Advanced option only.

Stop Bits – Forms part of the serial communication setting. For example, 8 data bits, 1 or 2 Stop bits no parity, serial communication. Ensure that the Stop bit settings are the same on the local and remote machines.

Retry – The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout – The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Use IEC Addressing – Instead of the old Modbus Addressing like 400001 it will display %MW0.

Moxa Cluster Support - Check this option when running on a MOXA virtual port in a cluster environment. It will allow to purge the port when in standby and in shadow mode, as the MOXA unit stops functioning in this mode. Data keeps filling the buffer seeing that the driver does not do any reading from the port and eventually becomes unresponsive.

Hard Purge (Using API) – Does a call to the COM PORT to purge all the data on it. (Preferred Method.)

Soft Purge (Read from Port) - Continue to read data from the port and just discard it.

3.1.3. Handshaking

RTS – (Disable/ Enable/Handshaking). (Default = Enable) Enable means line is always on, disable means line is always off, Handshaking means line is on when buffer is less than 1/2 full and off when buffer is over 3/4 full.

DTR – (Disable/Enable/Handshaking/Toggle). (Default = Enable) Enable means line is always on, disable means line is always off, Handshaking means driver will line is on when buffer is less than 1/2 full and off when buffer is over 3/4 full. Toggle means the line is high when bytes are being transmitted and low when the transmit buffer is empty.

CTS – (Enable/Disable). (Default = Enable) If enabled, the driver will wait for the line to be asserted before transmitting any characters.

DSR – (Enable/Disable). (Default = Disable) If enabled, the driver will ignore any characters received while the line is not asserted.

3.1.4. Network Latency/Delays

Between stations (network) - (Default = Inter-PLC delay) Use this option only if using a 422 network. A value of zero disables the option. The inter-PLC delay or latency is the delay in milliseconds between communicating between different slave devices on the network. This timing is needed to allow a slave to detect the end of the communication cycle.

Between Any Transaction (device) - Inter message delay. Use this option to have a delay between any two messages even from the same device.

Period in ms - The time in milliseconds to delay.

A Modbus slave interprets a 2.5 character time delay at the end of a message as a signal that communication has been broken off, and thus allows communication to be initiated by the master to a new slave. By default, the Adroit driver does not implement this delay.

To calculate the necessary time use the following formula, we have used 5 characters instead of 2.5 as this is always good engineering practice. 11 bits per byte have been used.

$1/\text{Baud} \times 11 \times 5$.

e.g. For a baud-rate of 2400, $(1/2400 \times 11 \times 5) = 0.022916$. or 23ms.

For a baud-rate of 19200, $(1/19200 \times 11 \times 5) = 0.002863$. or 3ms.



PLEASE NOTE

Rounding up of the interval to the nearest 20 ms is used.

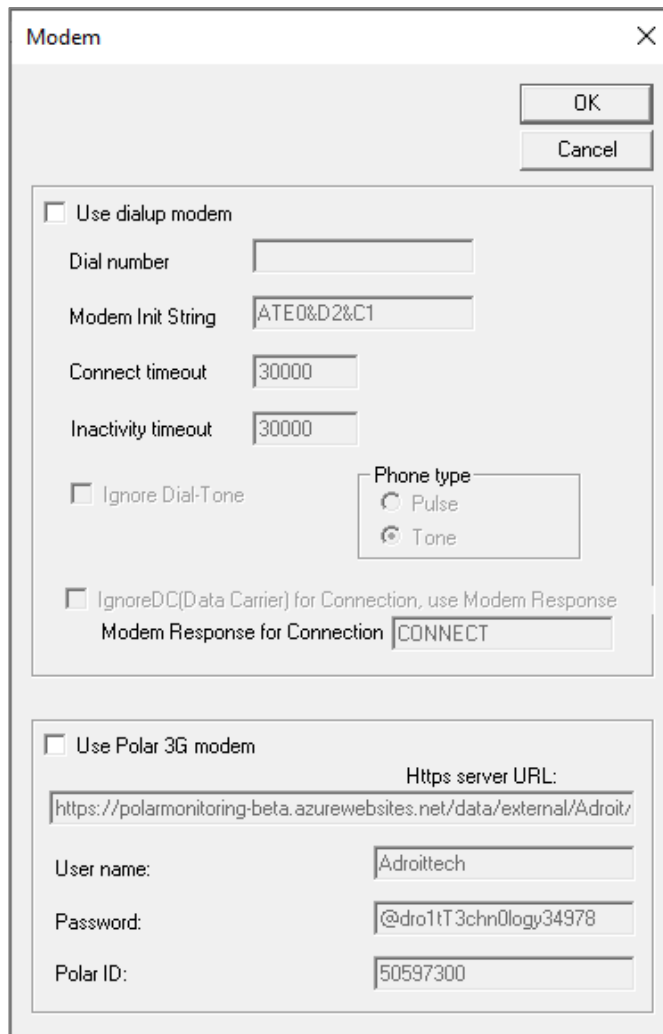
3.2. Modem Support

Going into the Modem dialog as shown below allows access to modem settings.

The modem support operates on the basis of dialling up an outstation whenever it needs to be scanned. When an outstation is scanned, the driver dials the outstation up and connects and performs any scanning of that station as may be required. Outstations sharing an identical phone number are treated as one connection, but when a station on another number gets scanned, the driver drops the line and dials up the new outstation.

The driver will "hold" the line for an inactivity period or until more scanning is done on that line, re-setting this period. After the inactivity period the driver will drop the line.

It is suggested that all devices normally be stopped, you can then start a device, causing it to scan and be dialled. Stop the device and then start another, in round-robin fashion. A mimic, graphic form or a recipe/script easily accomplishes this. Keep the device started for long enough to scan all your data. When the device is stopped, the tags go grey to indicates that they are offline. They become active again while on-line as you start and run each device.



3.2.1. Use Dialup Modem

– must be on for the driver to use the attached modem.

Dial number – Currently only supporting dialing out, you may use commas to indicate a pause e.g.: 0,1234567.

Connect timeout – After this time in milliseconds the driver stop waiting to connect.

Inactivity timeout – After this time, the driver will drop the line automatically to save dial-connecting cost and time. Make this value zero to have the driver drop the line immediately after each transaction regardless of the next transaction being on the same number.

Ignore Dial-Tone – Turn on to tell the driver to not wait for a dial-tone before dialling.

Pulse – Select if you use an analogue exchange.

Tone – Select on if you use a digital exchange.

Ignore DC – There is modems that is not wired for Data Carrier. This will Ignore the signal and rather check for Connection Response String entered below to determine a successful connection.

Modem Response for Connection – The connection string replied by the modem when an active connection to other modem is made.

3.2.2. POLAR 3G Modem Support

UsePolar-3G modem – (Default = false) enable the use of Polar 3g units instead of direct serial connections.

Https server URL – The connection send URL typically in the following format
<https://polarmonitoring-beta.azurewebsites.net/data/external/Adroit/SerialMessage>

Username – Username registered on the system.

Password – Password registered on the system.

Polar-ID – This unique device ID, this is not the Modbus address, but the two are technically paired.

Because the URL and subsequent transaction can take some time, a longer than usual timeout can occur, typically 30 seconds. Timing information is available on the datascope.

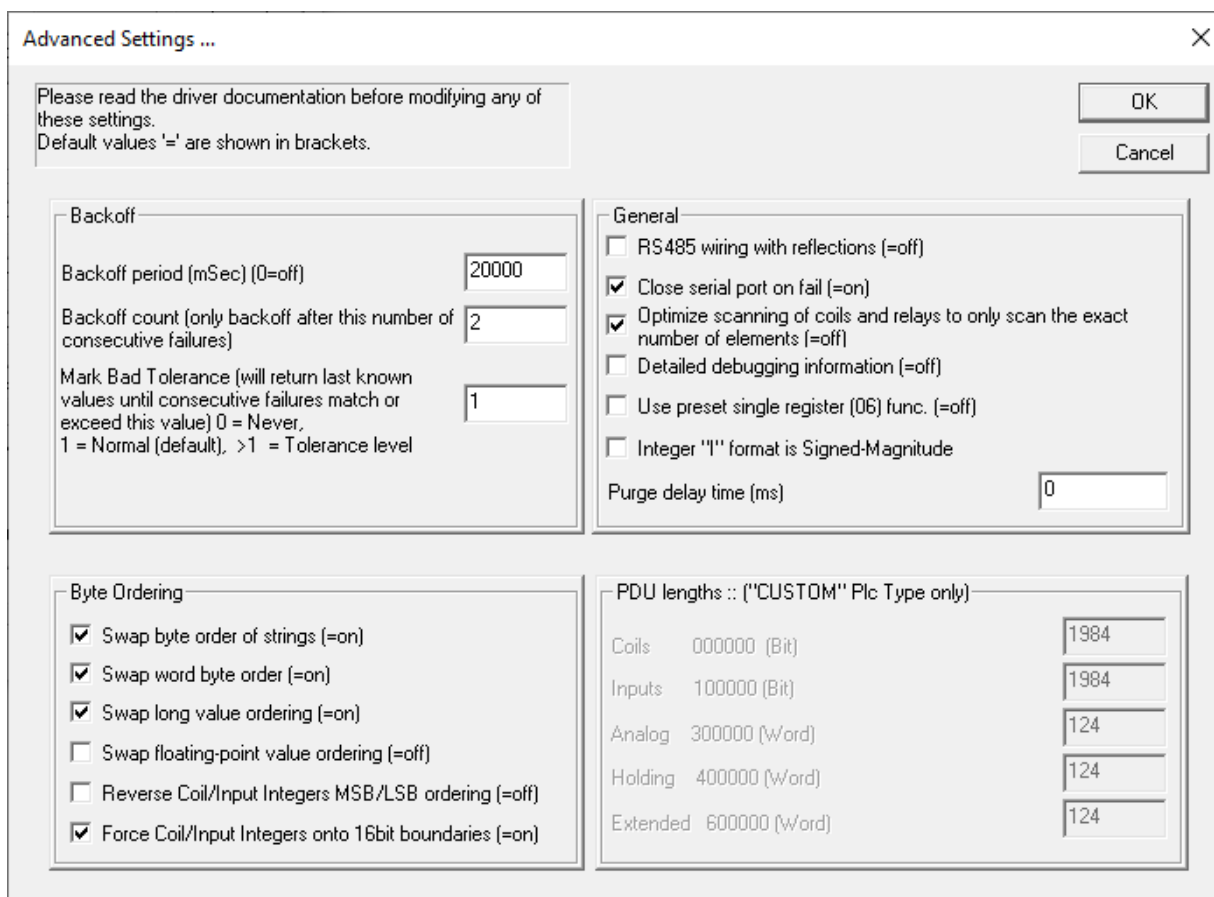
Polar-GSM devices make use of a secure HTTPS link to the cloud to communicate with the modem device that has a serial (RS-485 or RS-232) link to any Modbus serial (RTU-Binary) device. The SCADA machine will need to have access to the internet.

Some other settings are not applicable when using polar mode, and dialup modem mode is mutually exclusive.

Serial port is not required when using polar mode.

3.3. Advanced Settings

Many settings shown here are for advanced users only and were previously only accessible via the system registry.



3.3.1. Backoff

Backoff Period – (Default = 20000ms) A period in time(ms) to allow communication to settle when communications is bad and then trying again –(0 Disables function).

Backoff Count – (Default = 2) The number of failures to occur before the backoff is in use (0 Disables).

Mark Bad Tolerance – Will return last known values until consecutive failures match or exceed this value. Transaction attempts do not include retries, so if set to 2, and retries set to 3 then only after $2 \times (1 + 3) = 8$ attempts will the job be marked bad.

3.3.2. Byte Ordering

Swap floating-point value ordering – Allows the ordering of all floating point (suffix F) holding-register I/O to be word-swapped. Required for all genuine Modicon equipment. Checked means it is not swapping.

Reverse Modbus coils to integers – If Integers scanned to a coil address appear jumbled, use this setting to reverse the ordering.

Swap long value ordering – Allows the ordering of all long (suffix L) holding-register I/O to be word-swapped. Checked means it is not swapping.

Swap byte ordering of strings – Turn off to swap each byte of the string, e.g. “EHLL10” becomes “HELLO1” .

Swap word byte order – Turn on to swap each byte of the word.

3.3.3. General

RS485 wiring with reflections – (Advanced option) When using RS-485 the driver will receive and cater for message reflection which occurs because the RX circuit is enabled during the transmission of telegrams. (Used with RTS enabled and DTR enabled.) When using an external RS-232 to 485 converter this option may not be necessary.

Close serial port on fail – (Default = enabled) This setting, when enabled, will do a hard shut down and reinitialization of the communications Port. Some Ethernet to serial convertors hang under noisy conditions, so we compensate for this by only purging and not closing the port.

Optimize scanning of Coils and relays – turn on this option to prevent the driver from doing read-ahead optimization, and instead read only the required number of bits. By default, the driver will read all bits packed into a byte, hence the last coil or relay read is always at the end of an 8-bit multiple. This option prevents incorrect packing by some non-genuine RTUs.

Detailed debugging – (Default = Disabled) This setting will add additional information to the datascope.

Use Preset single register (06) func. – (Advanced option) (Default = disabled) Allow the driver to use function code 06Hex when writing to any single holding register types. If disabled, the driver uses function 10Hex which is more generic.

Integer “I” format is Signed_ magnitude – (Default = disabled) Integers will be expressed as signed magnitude if checked. Use, for example, 400001 I 0xFFFF will result in 0x8001 if checked else 0xFFFF if not.

Purge Timeout – (0 to disable) Will instruct the communication resource to idle for the specified period in milliseconds to wait for incoming noise/corrupted messages to clear from the port, this happens when the timeout is too short, and data is still being received.

3.3.4. PDU lengths (see section: *Protocol unit sizes*)

Outputs 000000 , Inputs 100000, Analogue 300000, Holding 400000 , Extended 600000 – Must be used in conjunction with the “CUSTOM” PLC type, set the appropriate lengths here. Length count is in items i.e., 000000, 100000 = 8 items per byte, 300000, 400000 = 2 bytes per item.

Bit types (Inputs 000000 , Outputs 100000) work optimally in multiples of 8 bits (to a byte), i.e. 16 will result in 2 data bytes returned in the data portion of the PDU. Some devices do not respond correctly if multiples are not 8.

Word types (Analog 300000, holding registers 400000, extended 600000) any number can be used. For 32-bit types; example “longs” or “floats” , then the PDU size will be rounded down.

3.3.5. Protocol Unit Sizes

If the PLC slave is not one of those in the list, and is emulating MODBUS, use this table to determine a match for the maximum telegram lengths supported by PLC. Selecting shorter, telegram lengths gives poor performance, but ensures greater compatibility with the device units. If PDU item lengths are known, use “Custom” but make sure you configure it correctly in the Advanced dialog.

The maximum number of items (Digital bits or 16-bit registers) or request sizes that can be communicated for the different Modicon PLC’s is shown below:

	84	884	184	384	484	584	984
000001/%Q	64	2000	800	800	512	2000	2000
100001/%I	64	2000	800	800	512	2000	2000
300001/%IW	4	125	100	100	32	125	125
400001/%MW	32	125	100	100	254	125	125
600001/%ME	32	125	100	100	254	125	125

In addition we define some customer PLC PDU lengths.

	256/16	256/64	GE	TSX Nano	Mod-Mux	Danfoss	Elpro
000001/%Q	256	256	256	128	64	0	48
100001/%I	256	256	256	128	64	0	48
300001/%IW	16	64	120	120	8	0	3
400001/%MW	16	64	120	120	8	9	3
600001/%ME	16	64	120	120	8	9	3

	Moscad	Gefran	Lovato	Electron	Custom
000001/%Q	70*8	20*8	0	10	@
100001/%I	70*8	20*8	0	10	@
300001/%IW	70	20	2	10	@
400001/%MW	70	20	2	10	@
600001/%ME	70	20	2	10	@

* The GE Multilin and Moscad models implement device-specific features, see ***Telemechanique and other support***.

@ You may specify PDU lengths based on vendor-supplied information see “Advanced Dialog”.

4. Supported Addresses

PLC data address types supported by the MODBUS Adroit Protocol driver:

4.1. Using Normal Modbus Type Addressing.

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Outputs	1	000001	065535	Digital	000001			
	1			Decimal		000001	000001	
	2			Signed Integer		000001 I	000001 I	
	3			BCD16		000001 B	000001 B	
Inputs	4	100001	165535	Digital	100001			
	4			Decimal		100001		
	5			Signed Integer		100001 I	100001 I	
	6			BCD16		100001 B	100001 B	
I/O Registers	7	300001	365535	Digital	300001.1			
	7			Decimal		300001	300001	
	8			Signed Integer		300001 I	300001 I	
	9			BCD16		300001 B	300001 B	
Holding Registers	10	400001	465535	Digital	400001.1			
	10			Decimal		400001	400001	
	11			Signed Integer		400001 I	400001 I	
	12			BCD16		400001 B	400001 B	
	13			Long Decimal		400001 L	400001 L	
	14			Float			400001 F	
	15			Pascal-String				400001 S20
Extended Registers	16	600000	698303	Char String				400001 C20
	17			Digital	600000.1			
	17			Decimal		600000	600000	
	18			Signed Integer		600000 I	600000 I	
	19			BCD16		600000 B	600000 B	
	20			Long Decimal		600000 L	600000 L	
	21			Float			600000 F	
	22			Pascal-String				600001 S20
	23			Char String				600000 C20

Note 1: Integer and Real writes are not supported to coils 000001 through to 165535 . These are read only mappings.

Note 2: PLC digital inputs are Read only.

Note 3: The string format for Pascal-style strings (Sxx) is as shown below:

REGISTER1= length

REGISTER2=bytes 1&2

REGISTER3=bytes 3&4

REGISTER4=bytes 5&6

REGISTER5=bytes 7&8 etc...

Hence the register 400001 if you scan 400001 s20 may not have more than 80 in it (the driver will crop the string at 20, since that is the max length you specified in the scan address). All bytes until a null byte or the length specified in the length WORD will get copied into the Adroit string tag scanned.

When we write the string, any unused bytes sent to the PLC will contain 0, e.g. Is you said hello, we would send the hex codes.

0005 (length)

4845 1

4C4C 2

4F00 3

0000 4

0000 5

0000 6

until...

0000 for register 10.

Non-Pascal-style flat ASCIIZ strings via the C address notation (Cxx) instead of the Sxx notation

Note 4: Digitals scanned to a holding register are mapped 16 through 1 (Modbus ordering).

Note 5: Extended Registers start from 600000 not 600001 like the other areas, so 600000 is Extended file 1, offset 0.

Note 6: The Address parser will be able to parse old 5 digit Modbus addresses and correctly convert these to the 6 digit address automatically, it can also read and convert automatically between the IEC and 6 digit Modbus addressing on the fly. Even mixed addressing will be interpreted correctly and stored into the correct format, depending on the device "Use IEC Addressing" in the setup dialogue.

4.2. Using IEC Addressing :

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Outputs	1	0	65535	Digital	%Q0			
	1			Decimal		%Q0	%Q0	
	2			Signed Integer		%Q0 I	%Q0 I	
	3			BCD16		%Q0 B	%Q0 B	
Inputs	4	0	65535	Digital	%I0			
	4			Decimal		%I0		
	5			Signed Integer		%I0 I	%I0 I	
	6			BCD16		%I0 B	%I0 B	
I/O Registers	7	0	65535	Digital	%IW0.1			
	7			Decimal		%IW0	%IW0	
	8			Signed Integer		%IW0 I	%IW0 I	
	9			BCD16		%IW0 B	%IW0 B	
Holding Registers	10	0	65535	Digital	%MW0.1			
	10			Decimal		%MW0	%MW0	
	11			Signed Integer		%MW0 I	%MW0 I	
	12			BCD16		%MW0 B	%MW0 B	
	13			Long Decimal		%MW0 L	%MW0 L	
	14			Float			%MW0 F	
	15			Pascal-String				%MW0 S20
	16			Char String				%MW0 C20
Extended Registers	17	0	98303	Digital	%ME0.1			
	17			Decimal		%ME0	%ME0	
	18			Signed Integer		%ME0 I	%ME0 I	
	19			BCD16		%ME0 B	%ME0 B	
	20			Long Decimal		%ME0 L	%ME0 L	
	21			Float			%ME0 F	
	22			Pascal-String				%ME0 S20
	23			Char String				%ME0 C20

5. Protocol Definition and Considerations

5.1. Diagnostic Sub-Functions

The function 08 sub-functions are supported as an extension to the driver's capabilities, and as such do not operate as standard scan-points, although they are licensed as such. Data returned from a function cannot be read into a tag at this point.

The diagnostic functions are advanced functions, and require special tag configuration for proper operation. To enable the function 08 support, set the registry value "enableFn08" to a non-zero value, this is the default. These functions are only supported in RTU mode, and not yet for ASCII operation.

5.1.1. Function 08 Supported Addresses

		Tag access	Min	Max	Boolean	Integer
Triggers	16	Write (rising edge trigger)	0	21	11 DT	
Values	17	Read (result of diagnostic request)	0	21		11 DD

The driver provides 22 digitals, which can be written to, a value of 1 (one) triggers the corresponding sub-function from 0 through 21. A value of zero is ignored, the driver only detects the rising edge, digitals should thus be configured as pulsed. The driver performs the sub-function when triggered and stores the communications result in a further list of 22 Integer slots. These can be read using multistates or integers. Where applicable, they may also be written to, this will in certain cases affect the way that the trigger will work for the corresponding function.

5.1.2. Startup Function

Facility is provided to automatically trigger one function only, whenever the device is started, as-if the relevant digital was triggered, the start-up feature is disabled by default. To enable this feature use regedit.exe to enter a valid code in the value "startupFn08code". An invalid code, (for example, 65525) simply disables the start-up feature and no further warnings are issued.



PLEASE NOTE

Not all devices support the full range of sub-functions, the driver makes no attempt to validate the model for a particular function's compatibility. Use of certain diagnostic sub-functions is thus at own risk.

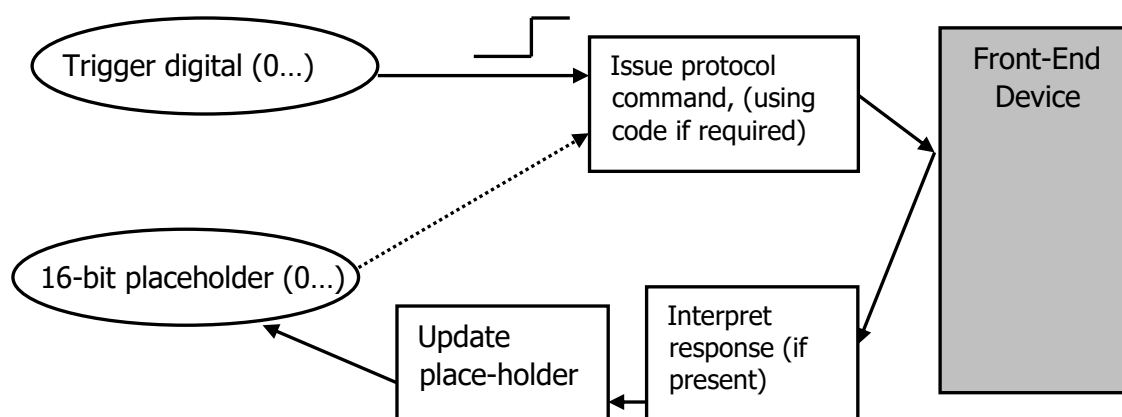
Certain functions do not provide responses, for those that are, the driver copies the response data field to the relevant integer placeholder. The value is then read on the next scan. For functions that require data, the same placeholder may be written to, to provide the necessary data. The data for sub-function 00 can be pre-set to a value by entering it into the registry, see *comms recovery* below. The internal read and write locations of the place-holder are separate. Reading that location returns any data returned from the last time the function was transmitted to the FED, or zero. Writing to the location transfers the user data in the place-holder to the FED.

5.1.3. Comms Recovery

After a communications failure, the driver will automatically issue the start-up function if it is a valid one. This will accommodate a FED that gets re-set and allow adroit to re-initiate communications to the device smoothly. If the start-up function is enabled, the driver will continue to use the start codes to try and re-initiate communications, until a scan succeeds. Any communications fault will force the driver to re-init. See the registry entries section for the data to send in the start-up function.

5.1.4. Tag Configuration

Digitals configured for triggers must be **output enabled** and **input disabled**. Integers configured for placeholders only need to be output enabled if applicable. The engineer need not configure a corresponding placeholder for every trigger.



5.1.5. Function Reference

The function codes and a brief explanation of each follow, for more detail see Modicon Modbus Protocol Reference Guide – PI-MBUS-300:

Sub-function Code		Description
0	00 00	Return query data (echo function)
1	00 01	Restart communications option. Restarts and initializes slave's peripheral port. This function also brings a slave out of listen only mode. (Option) data code 0xFF00 clears the slaves comms event log, data code 0x0000 leaves log as-is.
2	00 02	Return diagnostics register. (returns the slave's 16-bit diagnostic register, see Modbus manual for bit meanings)
3	00 03	Change ASCII delimiter (Option) . The high byte of the 16-bit data code corresponds to the ASCII character to use.
4	00 04	Force listen-only mode. While in this mode watchdog will expire. Use sub-code 01 to get out of this mode.
5-9		Reserved
10	00 0A	Clear counters and diagnostics register.
11	00 0B	Return bus message count.
12	00 0C	Return bus communication error count.
13	00 0D	Return bus exception error count.
14	00 0E	Return slave message count.
15	00 0F	Return slave no-response count.
16	00 10	Return slave NAK count.
17	00 11	Return slave busy count.
18	00 12	Return BUS character overrun count.
19	00 13	Return IOP overrun count (884 specific)
20	00 14	Clear overrun counter and flag.

21	00 15	Get/Clear Modbus Plus statistics. A value of 03 in the data code retrieves the statistics, and code 04 clears the statistics. Only the first 16 bits of the returned data for this function are processed and placed in the data code slot.
----	-------	---

5.1.6. Function 08 Registry Entries

The 3 registry entry values can be found in the "Primary" key for each device. e.g.

"HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\MODBUSI\XMODBUS\Primary"

DWORD **startupFn08code** = FFFF

DWORD **enableFn08** = 1

DWORD **startupSF00data** = AA55

By changing the "enableFn08" value to zero, reading and writing of the diagnostics points can be disabled for security. The values in "Secondary" are ignored if they are different. If the diagnostics functions have been disabled, the driver addressing options message will no longer show the explanation regarding the diagnostic scan configuration. The StartupSF00data value can be used to set an initial value for the sub-function 00 data. e.g.: 0xaa55.

ABB Specific

For proper communication to ABB AC400 series, the diagnostic functions must be enabled, use the following registry settings. Settings tested with the model AC 450.

DWORD **startupFn08code** = 0

DWORD **enableFn08** = 1

DWORD **startupSF00data** = AA55

5.2. Broadcast Support

Broadcast support is achieved by configuring an additional device for station #0. The Modbus protocol uses a station ID of zero to perform broadcasts. The driver will return zero on scan of any coils/registers on address zero, and when writing, will not wait for any response from the slave(s). If you do not want the database to be updated with the zero, you must scan-inhibit the tag while output enabling it.

All slaves will interpret the command if it is valid. Slaves do not reply to broadcast commands and only support the following commands in broadcast mode, commands not used by the driver are not shown.

Broadcast commands

Code (HEX)	Description
05	Write single coil
10	Preset/Write multiple holding registers

For a full list of commands see *Message Protocol* below:

5.3. Telemechanique and other support

The Telemechanique Premium, Micro and Nano V3 operate in the same way as the original Modicon devices. Differences encountered in cabling are covered here.

Using TSX SCP CU 4030

TSX SCP 114 (slave)		RS 484->232 Converter		PC 9Pin
White/blue	_____	A TD	_____	2
Blue	_____	B RD	_____	3
			┌	4
			┌	6
			┌	7
			┌	8

Using TSX SCP CM 4030

TSX SCP 114 (slave/master)		RS 484->232 Converter
Orange	┌	A
White/gr	┌	B
Green	┌	
White/or	┌	
Brown	┌	

5.3.1. GE Multilin Features

Support for the GE Multilin affects only digital coils & relays allows for digital off signals to not be transmitted, nor will digital outputs be read in, but instead will always assume a zero value.

5.3.2. SuperCon Power Meters

Use PLC model 384, 9600 baud (selectable via menu on the meter), 8 data bits and parity set to none. The analog values in the meter must be scanned as float, in the conventional way, I.e. "400015 F" will be the analog scan-address of the 1st phase voltage. Use an analog with zero or no scaling.

5.3.3. MOSCAD

When selecting Moscad as the PLC model in order to communicate to a MOSCAD RTU, the driver will build transactions based on the Table #, Row# and Column # rules for Analog and Holding registers. Holding Register addresses are in a long format allowing access to 56635 registers, register addresses are 400001 thru 465535.

5.3.4. ABB Controllers

Require some setting up of Function 8 diagnostics, see the section *diagnostic sub functions* for detail

5.4. Advanced Registry Key Settings

The following settings are not configurable in the normal fashion and are accessible via the registry. These settings now appear in the [Advanced dialog](#).

These value(s) can be found in the following key.

"HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\MODBUSI\MYDEVICE\Primary"

enableFn08

startupFn08Code

startupSF00data, these registry values are covered in *diagnostic sub functions*.

These value(s) following can be found in the following key.

"HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\MODBUSI

SoftRedundancy: (Default = 0) If the value does not exist. **This setting affects all Modbus serial driver devices globally.** When turned on, individual devices do their own port switching when used in a Primary/secondary pair. The default behaviour causes all devices to switch at the same time, and assumes that when primary or secondary fails, that no devices on that channel are accessible at all. Setting this value to 1 allows a network to be looped between the Primary and back to the Secondary port and survive any *single* network break.

5.5. Message Protocol

5.5.1. Coils 00001- 065535 (%Q0-%Q65535)

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	01	00	13	00	10	xx

Read response:

Slave Address	Function	Byte Count	Data	Data	CRC
2	01	02	02	0F	xx

Write request:

Slave Address	Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo	CRC
2	05	00	13	FF	00	xx

Write response:

Slave Address	Function	Coil Addr Hi	Coil Addr Lo	Force Data Hi	Force Data Lo	CRC
2	05	00	13	FF	00	xx

5.5.2. Inputs 10001- 165535 (%I0-%I65535)

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	02	00	13	00	10	xx

Read response:

Slave Address	Function	Byte Count	Data	Data	CRC
2	02	02	02	0F	xx

Write request:

Not supported

5.5.3. Input Registers 30001- 365535 (%IW0-%IW65535)

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	04	00	13	00	1	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	04	02	02	0F	xx

Write request:

Not supported

5.5.4. Holding Registers 400001- 465535 (%MW0-%MW65535)

Read request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Points Hi	No Points Lo	CRC
2	03	00	13	00	1	xx

Read response:

Slave Address	Function	Byte Count	Data Hi	Data Lo	CRC
2	03	02	02	0F	xx

Write request:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Registers Hi	No Registers Lo	Byte Count
2	10	00	13	FF	02	xx

Data Hi	Data Lo	Data Hi	Data Lo	CRC
2	10	00	13	xx

Write response:

Slave Address	Function	Start Addr Hi	Start Addr Lo	No Registers Hi	No Registers Lo	CRC
2	10	00	13	00	02	xx

Write request for 16-bit types if function 06 enabled:

Slave Address	Function	Start Addr Hi	Start Addr Lo	Data Hi	Data Lo	CRC
2	06	00	13	2	10	xx

Write (06) response:

Slave Address	Function	Data Hi	Data Lo	CRC
2	06	2	10	xx

5.5.5. Diagnostic 0x08 functions

Typical write request:

Slave Address	Function	Sub-code Hi	Sub-code Lo	Query-code Hi	Query-code Lo	CRC
2	08	00	01	00	00	xx

Write response:

Slave Address	Function	Sub-code Hi	Sub-code Lo	Data-code Hi	Data-code Lo	CRC
2	08	00	01	00	00	xx

Note: Sub-function 21 (0x15) returns 54 words of response data; see function code descriptions.

5.5.6. Extended Registers 600000- 698303 (NB 600000 is valid) (%ME0-%ME98303)

Read request:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word count Lo
14	07	06	00	01	00	02	00	01

Read response:

Function	Overall Count	Byte Count	Register Area	Data Hi	Data Lo
14	04	03	06	12	34

Exception error response: (see table below for error definition)

Function	Code
94	02

Example: Read 1 register at reference 1,2 (File 1 offset 2 : 600001) resulting in value 1234 hex
(Note: File 1 ,Offset 0 = 600000)

Write multiple registers 09 request:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word Count Lo	Data hi	Data Lo
15	09	06	00	01	00	02	00	01	12	34

Write response:

Function	Remainder Count	Register Area	File Addr Hi	File Addr Lo	Offset Addr Hi	Offset Addr Lo	Word count Hi	Word Count Lo	Data hi	Data Lo
15	09	06	00	01	00	02	00	01	12	34

Exception error response: (see table below for error definition)

Function	Code
95	02

Example: Write 1 register at reference 1,2 (File 1 Offset 2 : 600001) to 1234 hex

Note: File 1 ,Offset 0 = 600000

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. PLC Exception Codes

These codes are returned by the PLC. Some action either in scan configuration, driver setup parameters or hardware configuration may be needed to rectify the problem.

Hex code	Meaning	Description
0x0	No error	
0x1	Illegal Function	The function code received in the query is not an allowable action for the slave.
0x2	Illegal Data Address	The data address received is not an allowable address for the slave.
0x3	Illegal Data value	A value contained in the query data field is not an allowable value for the slave.
0x4	Slave Device failure	Unrecoverable error occurred in the slave while processing the request.
0x5	Acknowledge	This response is returned if the slave is processing the request but will take a long time; this is to prevent the master from timing out.
0x6	Slave device busy	Re-transmit the command later when the slave is not busy.
0x7	Negative acknowledge	Slave cannot perform a programming function requested.
0x8	Memory Parity error	The slave may require service. This error may occur when a request is made to read extended memory and a parity error is detected in the memory.

6.4. Trouble Shooting

If you are using a clone and have problems scanning floating-point or long (32 bit) values see the sub-section *Advanced settings dialog*.

If using a network of PLCs and experience intermittent communication errors see the net-latency option in the *Configuration Dialog* section.

If the datascope screen shows that the driver is receiving an echo of the transmit string, try restarting with the **RS485 wiring with reflection** setup dialog option on/enabled.

If the driver fails to initialize at start-up, but event logs a “Mode” error, this may be due to the selection of a baud-rate not supported by your PC hardware. It may also occur for an illegal port setting combination for your hardware is used.

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
2.0	09-Jan-2023	K. Robinson	Grammar and formatting changes.	D. Jordaan
3.0				
4.0				
5.0				
6.0				
7.0				

7.2. DLL Revision

Rev.	Date	Changes
98	11-May 2021	Added settle time when using polar and the driver experiences comms fails Added more detailed debugging info to the datascope
99	13-Jul 2022	Driver could have high cpu load , fixed